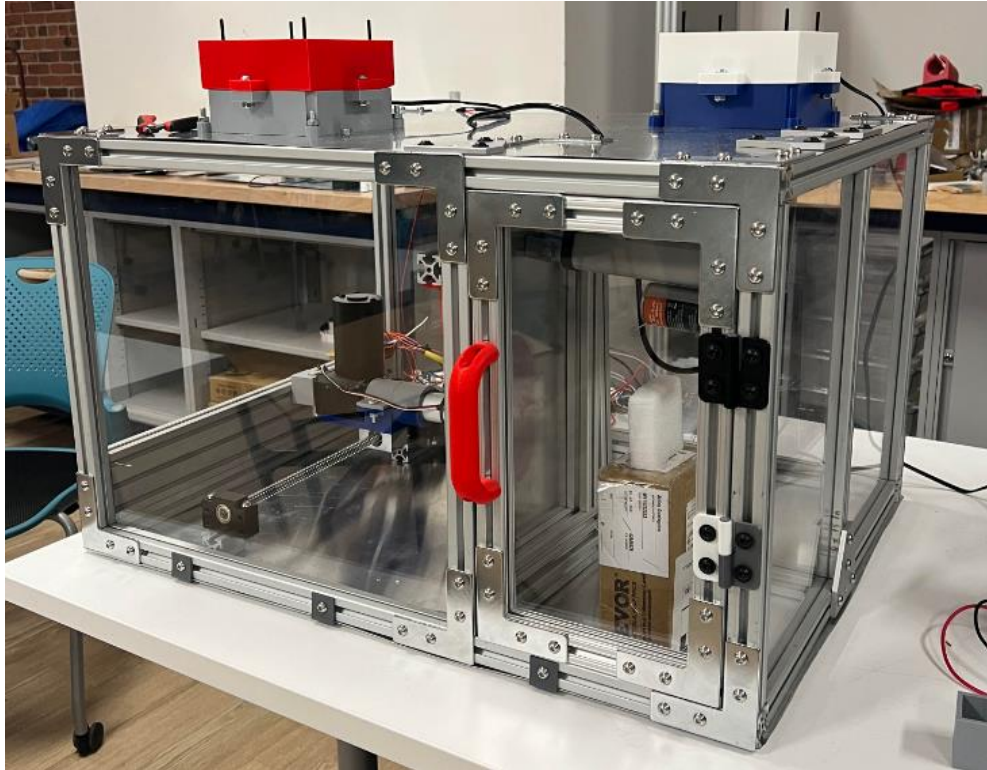




Fairfield University

School of Engineering & Computing

ASML



Project ISO: Tabletop Cleanroom

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Advisor: Dr. Andy Judge



Background

ASML - Advanced Semiconductor Materials Lithography

- ASML builds EUV/DUV lithography systems that pattern circuits at the nanometer scale
- Powers chips used in phones, laptops, and AI hardware

Wilton, CT

- Develops high precision optics, light source, and advanced modules





What's a Cleanroom

- Controlled environment to minimize particle contamination, dust, vibration, and temperature changes
- Essential for nanoscale accuracy in chip production

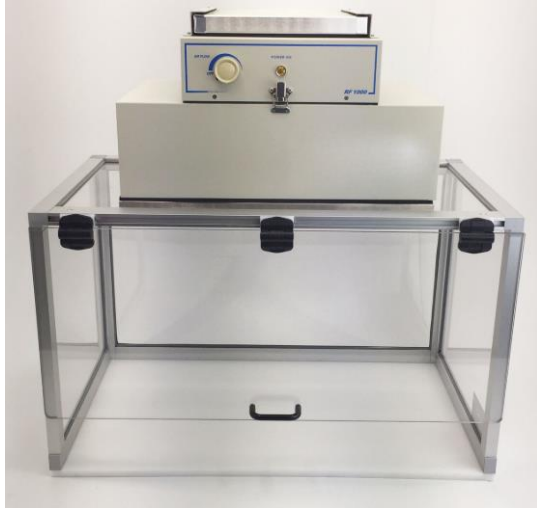




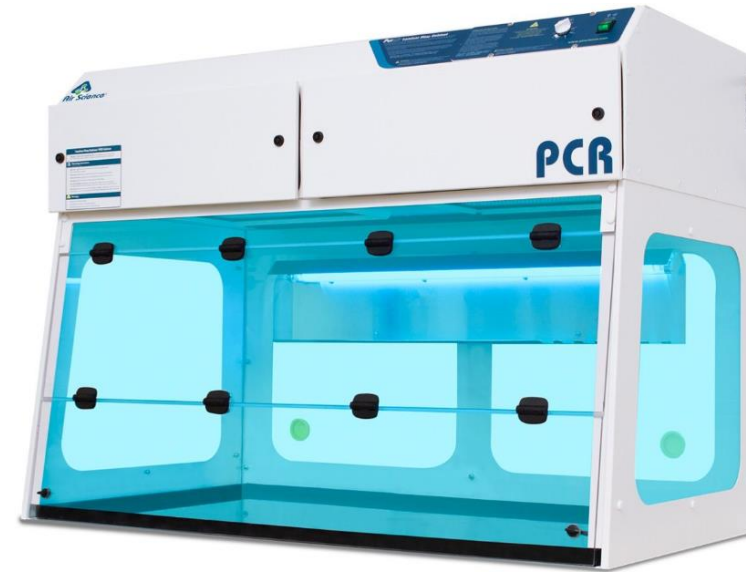
Current Market

Tabletop Cleanrooms

- Portable, compact clean workspace enclosures
- ~\$3,000 + not fully sealed
- Manual hand use = particles = risk of contamination



Cleanroom World



Polymerase Chain Reaction (PCR) Workstation

Project Overview

Problem to be solved

- Traditional cleanrooms - large and expensive
- Not easily accessible- limit research mobility
- Tabletop cleanrooms- expensive and manual



Objectives



Solution

- Develop a portable, cost-effective tabletop cleanroom (\$1k)
- Provide Fairfield University with cleanroom research capabilities
- Establish a platform for future partnerships with organizations like ASML

For Fairfield labs:

- Energy Transfer Lab, Thermal and Fluids Lab, Heat Transfer, Fluid Mechanics



Engineering Standards

- **ISO** - cleanroom classification, environmental testing, mechanical tolerances, and system verification
- **IEC** - electrical safety
- **NIST-Traceable Calibration** - sensor accuracy, repeatability, and measurement traceability
- **IEEE** - telemetry system, communication protocols

Design Specification

Enclosure

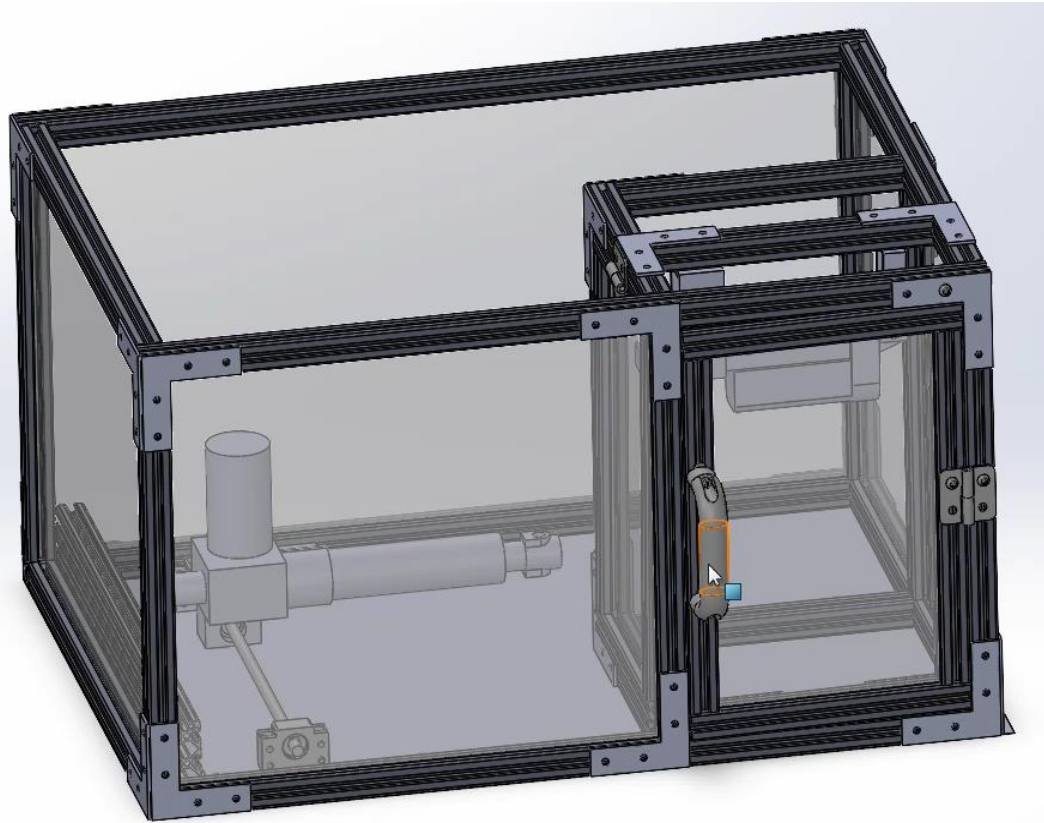
- Easy to assembly and cost effective
- Fully automated mechatronics

Cleanliness and Airflow

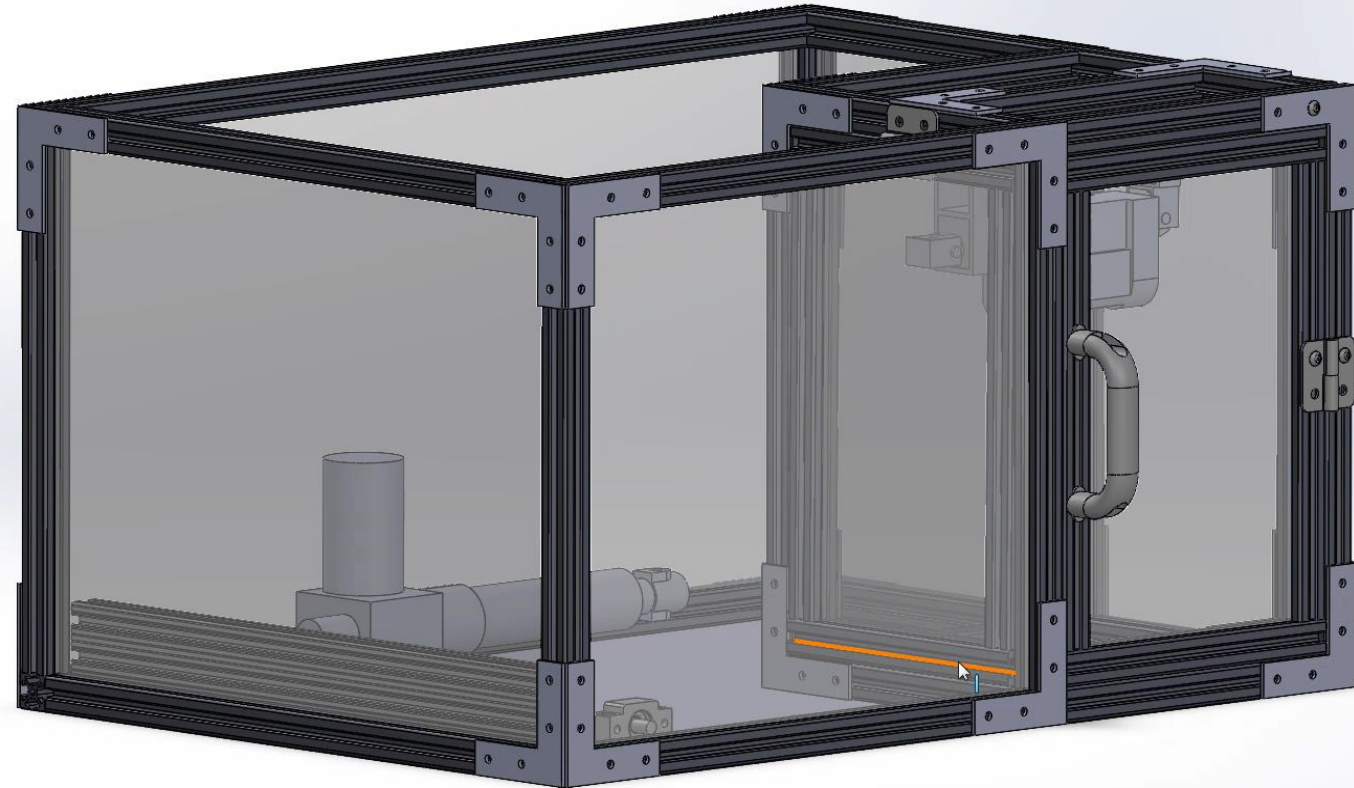
- ISO 6 Target : $\leq 35,000$ particles / $\geq 0.5 \mu\text{m}/\text{m}^3$
- Downward laminar airflow
- Air-change rate: > 180 ACH
- Positive pressure: $+5\text{-}10$ Pa inside chamber



Box Design Specifications- 29" x 22" x 17"



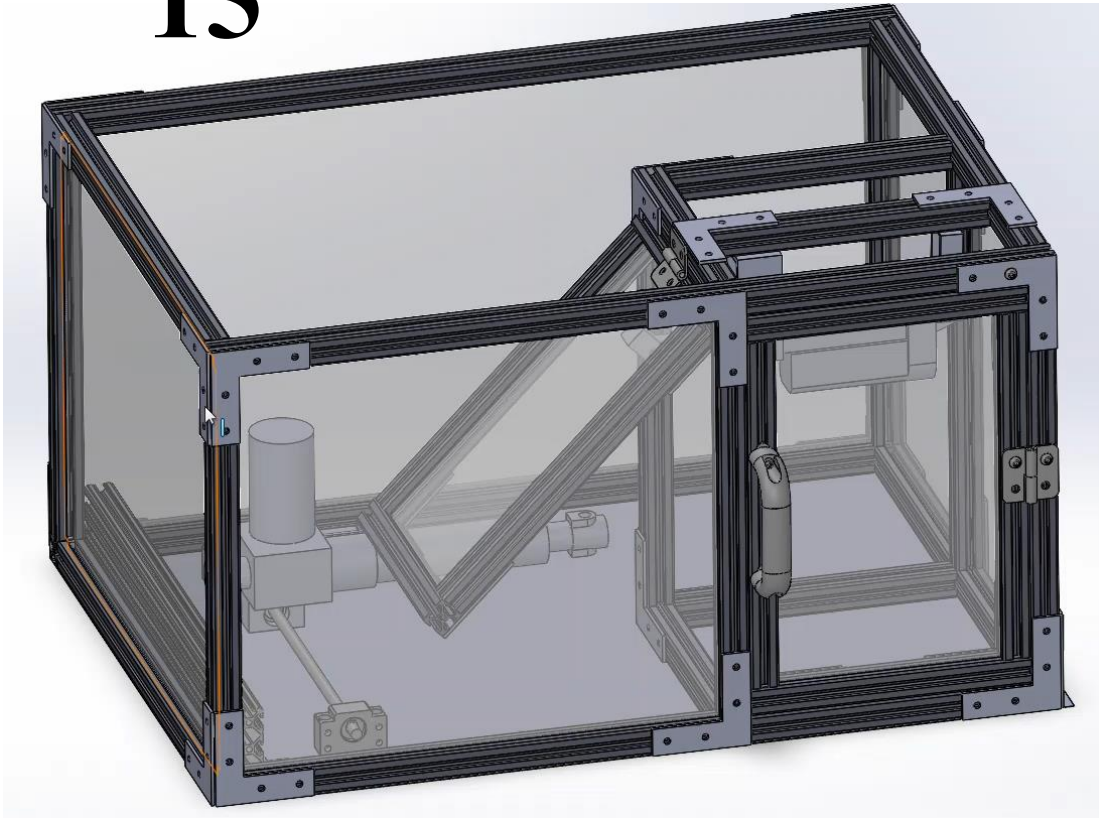
Manual Door



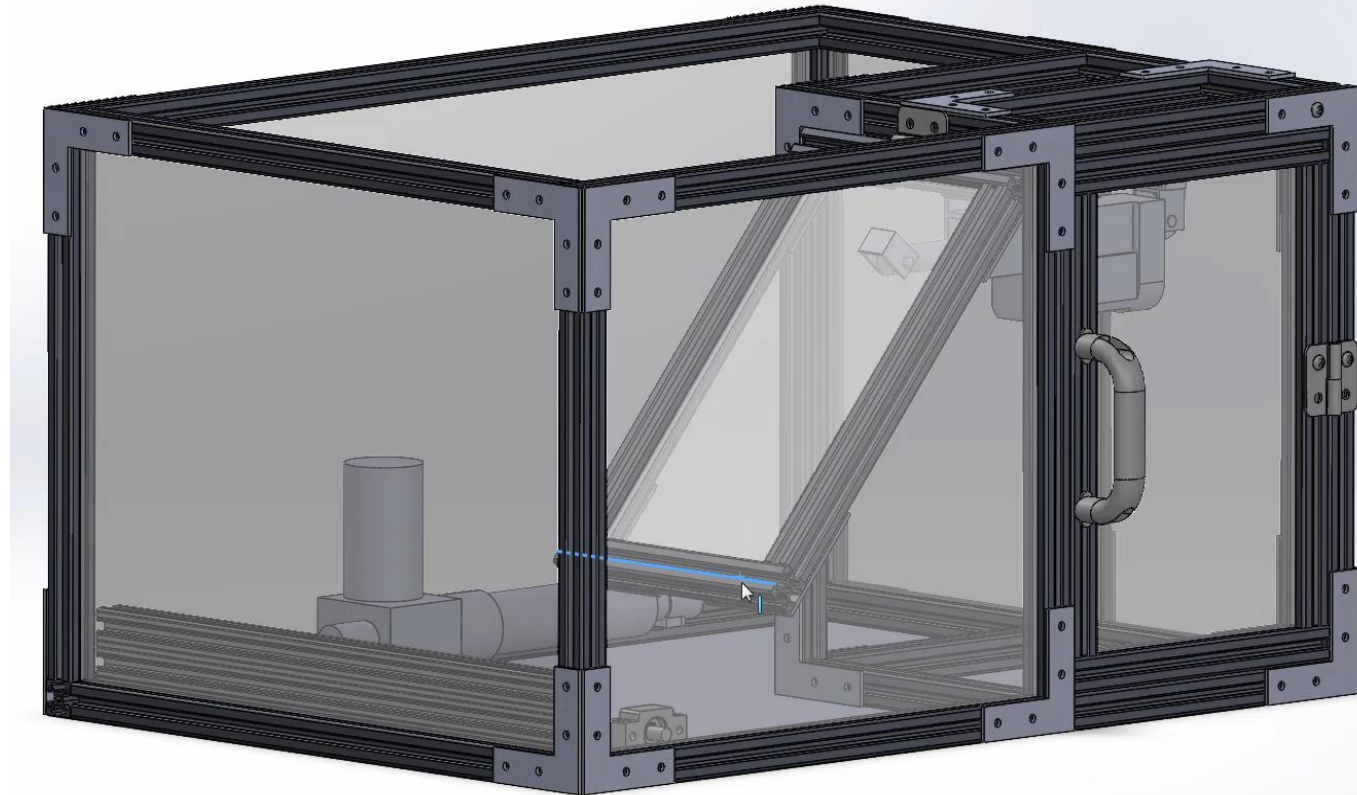
Automated Door



Door Design Specifications – 10 $\frac{3}{4}$ ” x 11” x 15”



Ball Screw and actuator



Automated Door

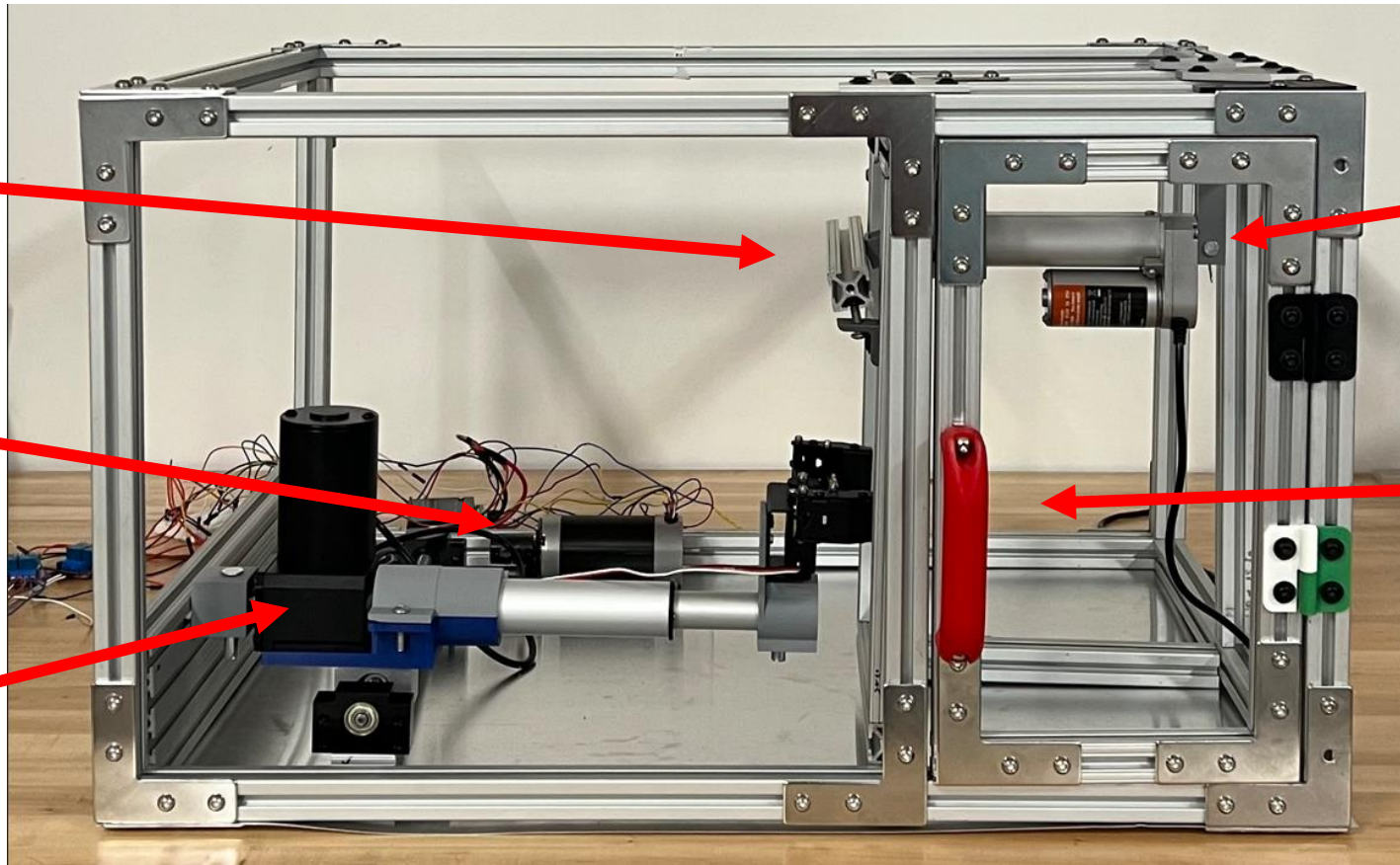


Final Product Overview

Automated
Door

Ball Screw
Motor

Ball screw
and actuator



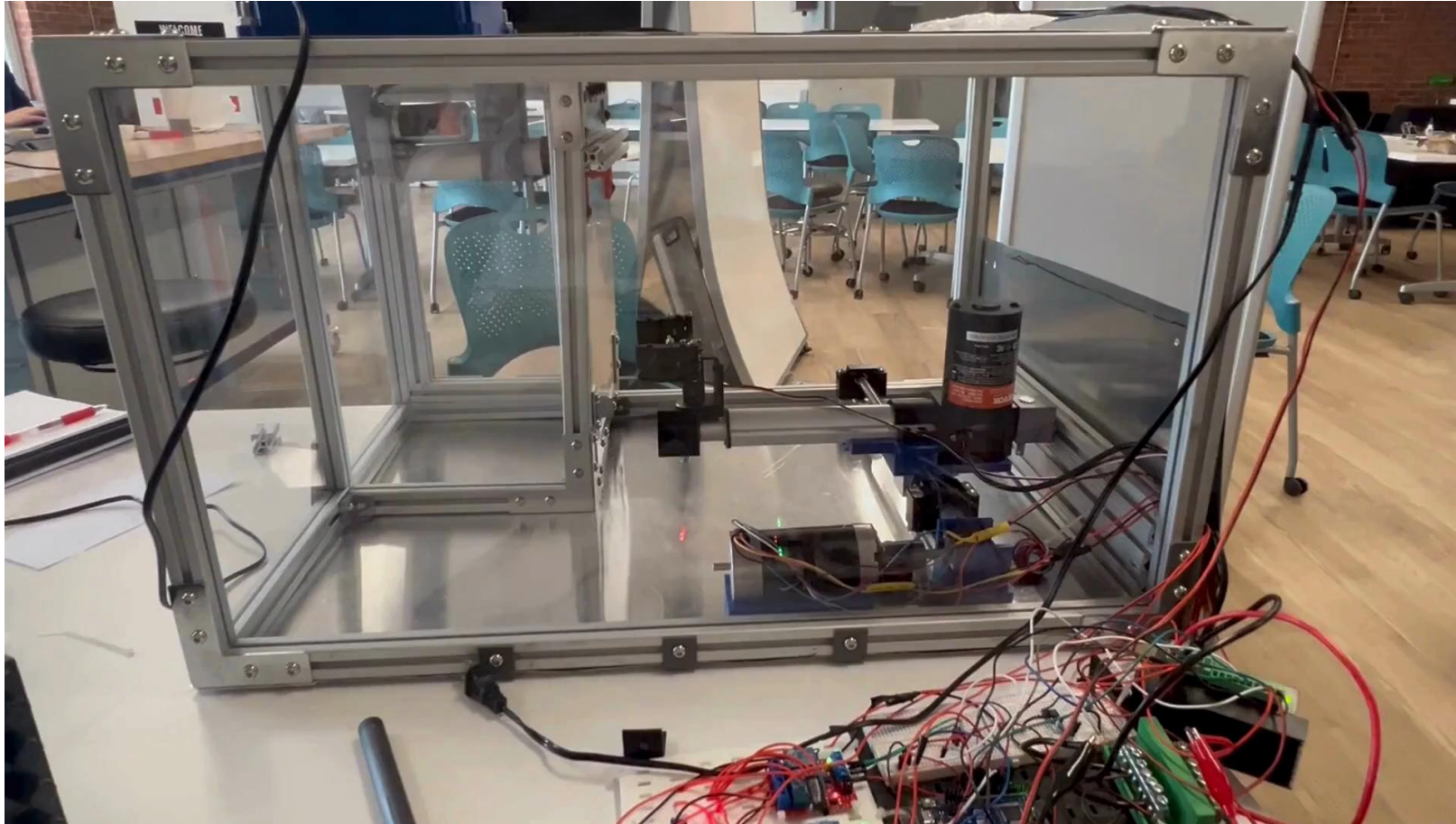
Door
Actuator

Manual
Door



System Demonstration

4x Speed



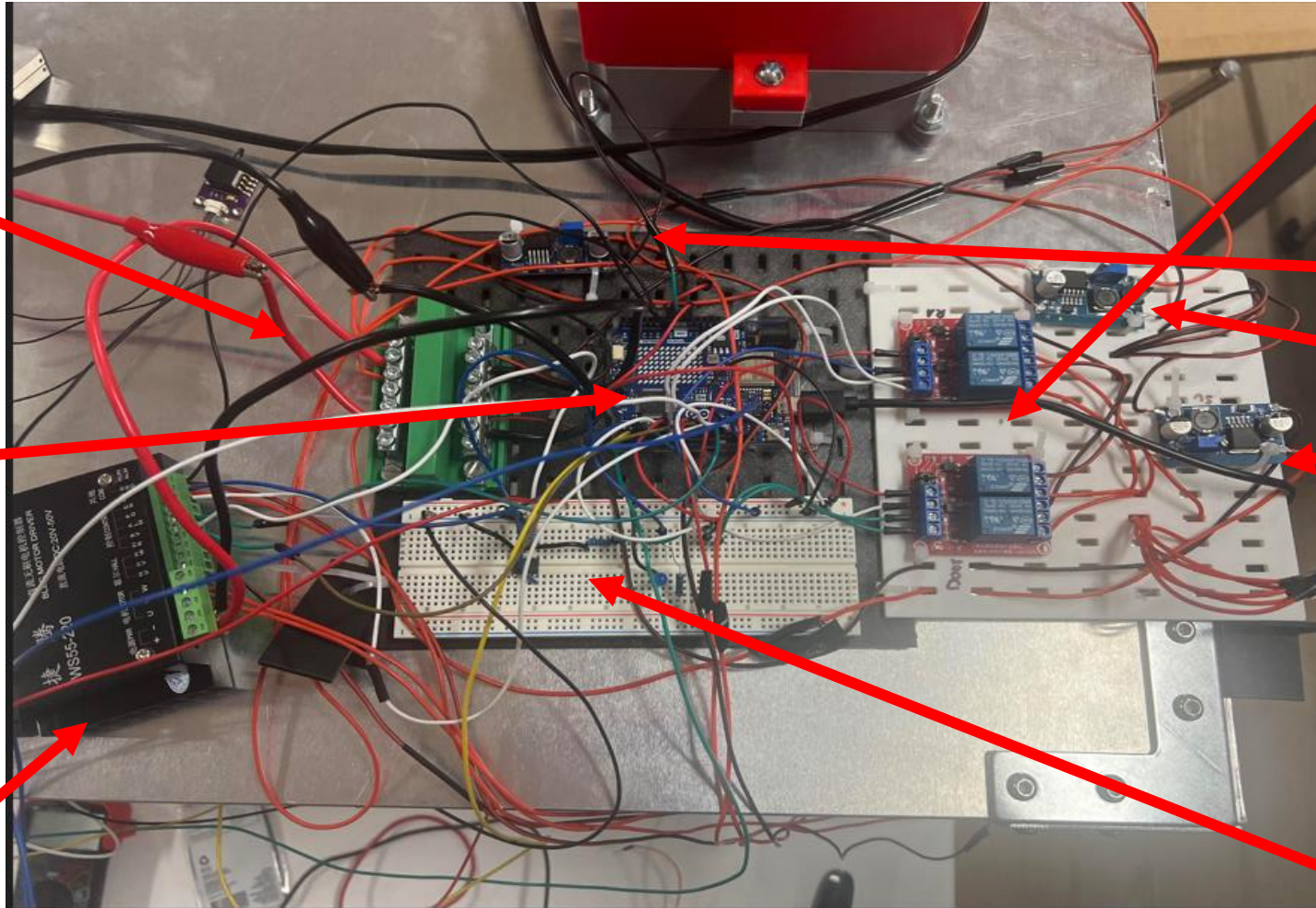


System Architecture- Mechatronics

Power supply

Arduino Uno R4

WS55-220 Motor Driver



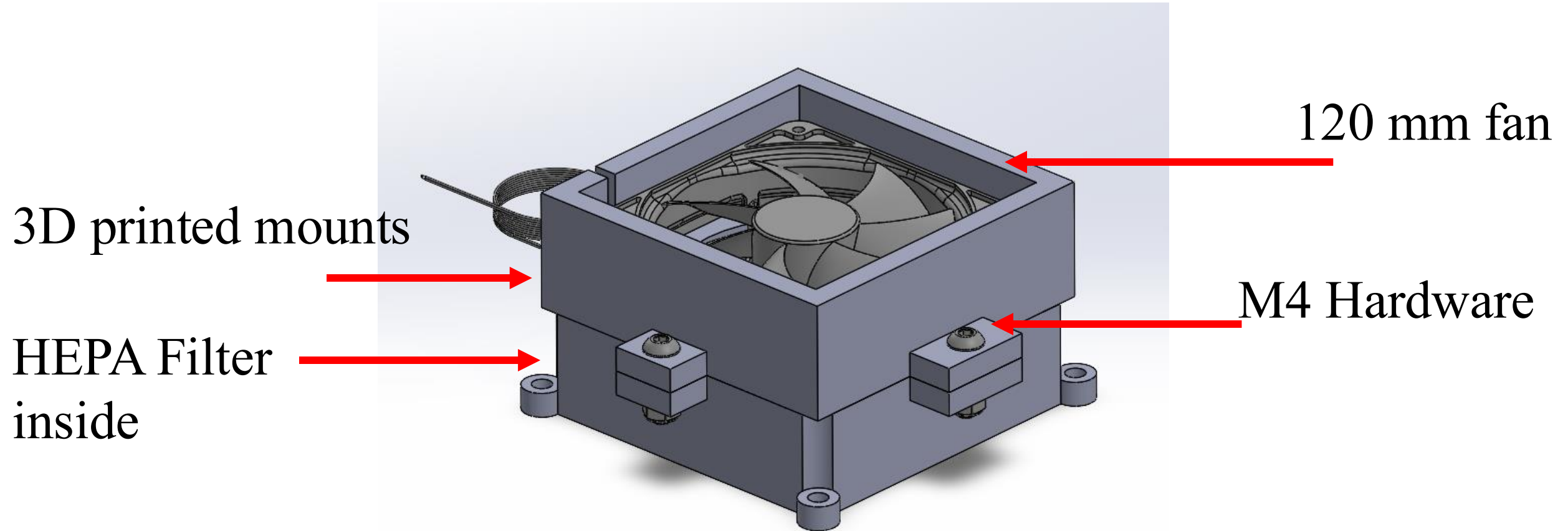
2 Channel Relays

Buck Converters

Breadboard/
RC Filter



System Architecture- Airflow





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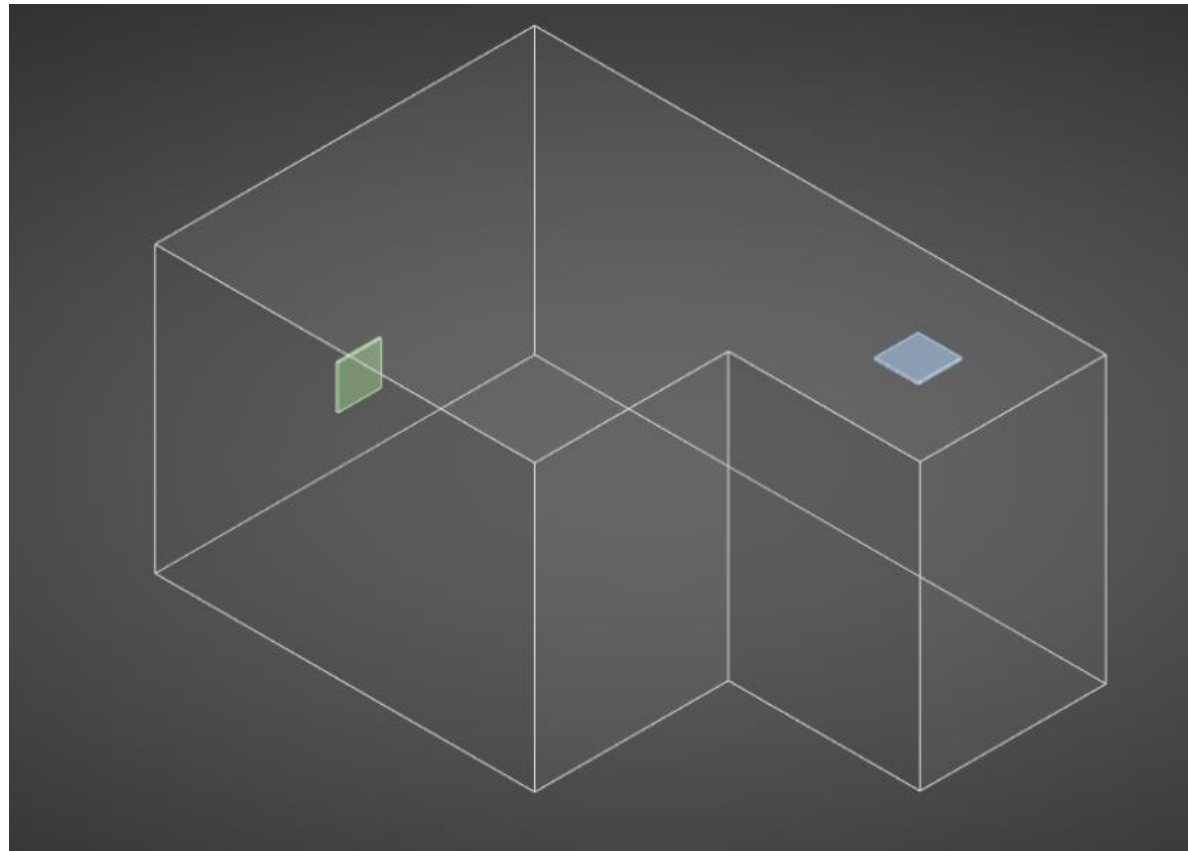
ASML

System Architecture- Airflow

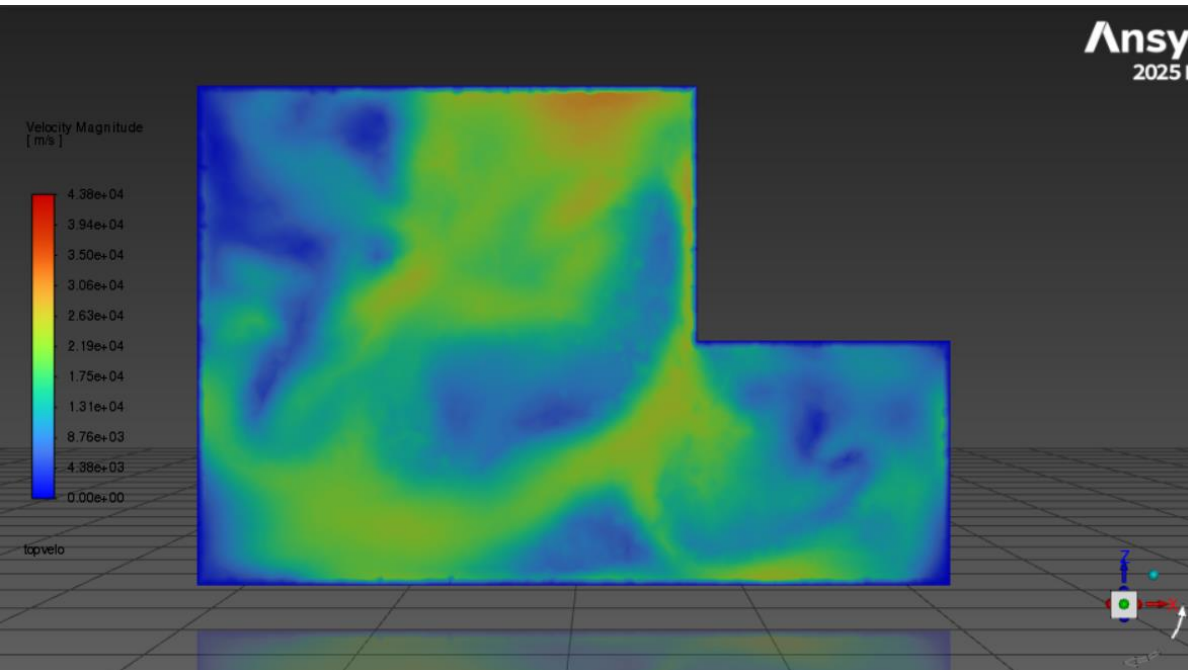




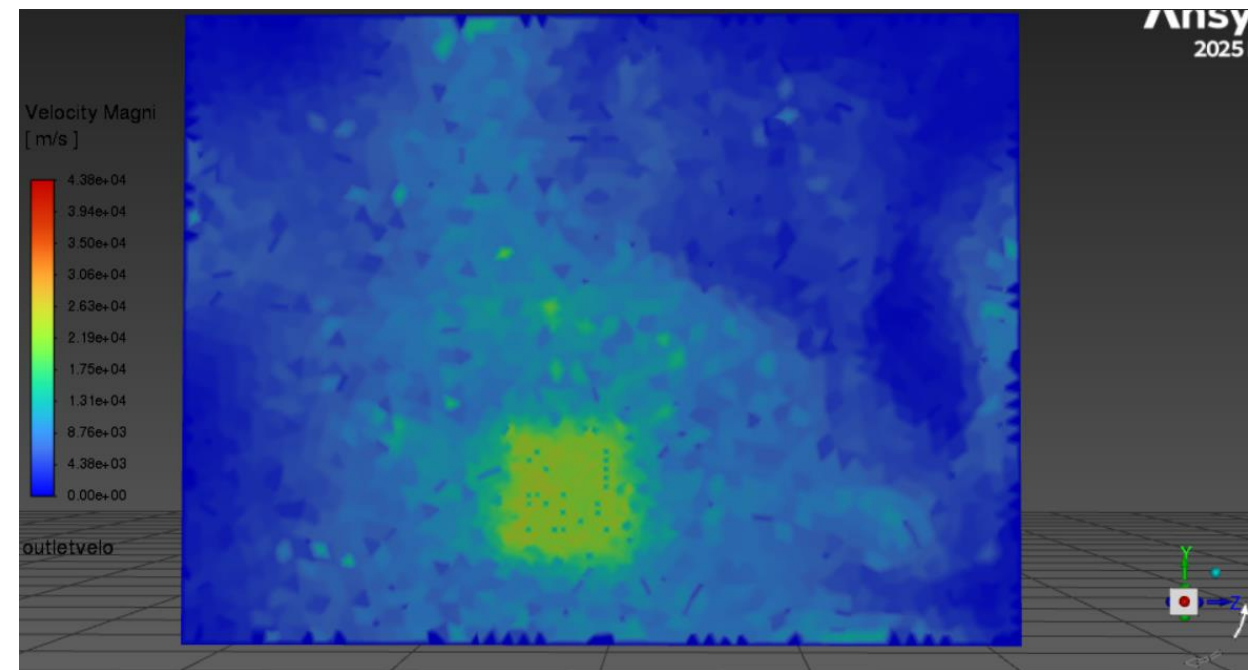
Work Completed - CFD



Work Completed - CFD



Top View (inlet)



Side View (exhaust)

Low velocity

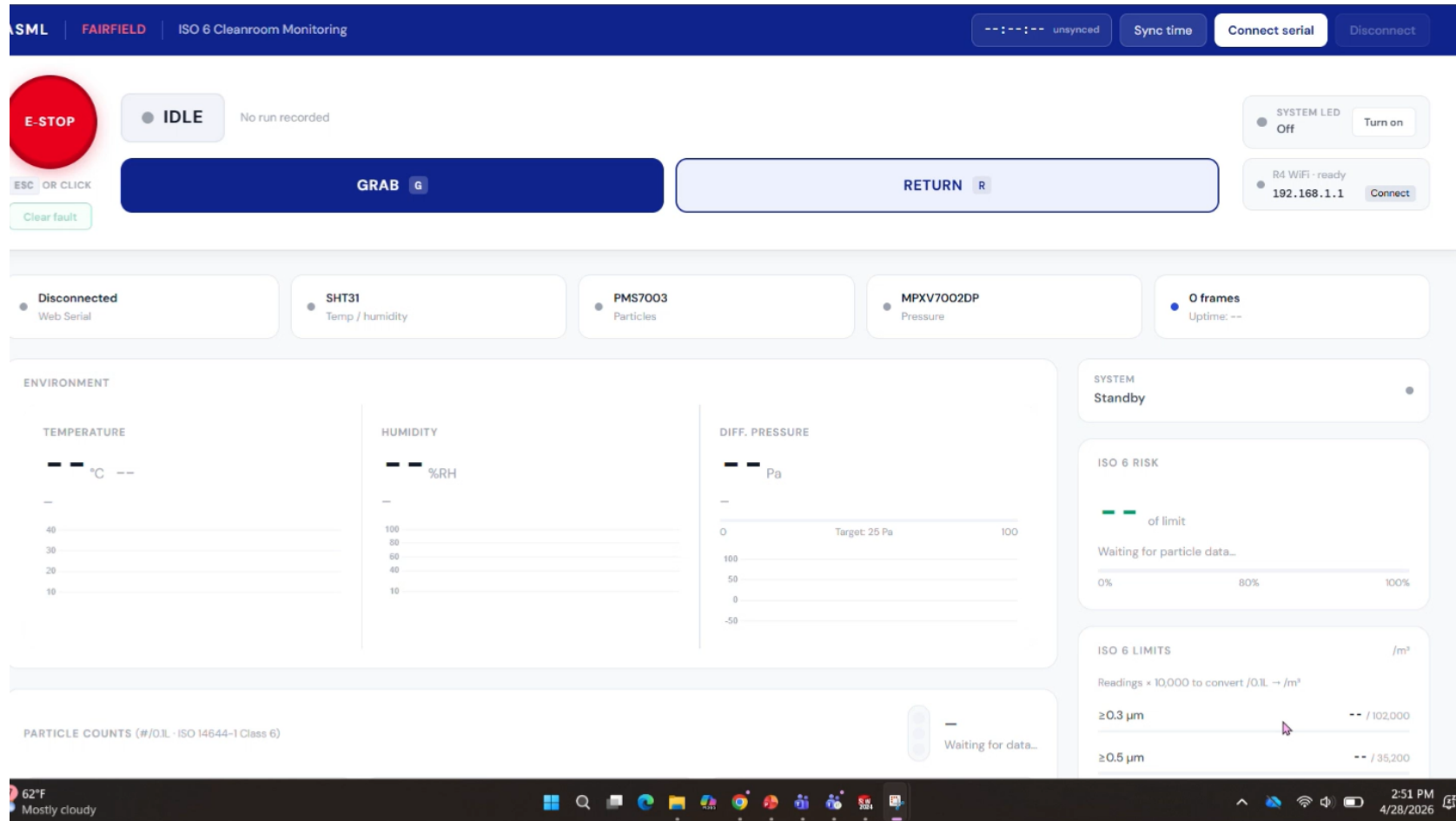


High velocity



Performance

Dashboard automated control system and telemetry monitoring



Challenges, Testing, and Validation

Enclosure

- 8020 and disassembly

Dashboard

- Troubleshoot dashboard with HTML to connect wirelessly

Mechatronics

- Arduino issues...
- Motor troubleshoot
- Voltage and Amperage budget

Airflow

- Fan speed

Telemetry

- **Accuracy**
 - Dashboard correctly classifies environment *as ISO Class 6*
- **End-to-End**
 - Sensor → Arduino → Web flow with zero data loss



Future Work

- Integrate box with current ASML particle counter
- Utilize modern cameras to photograph cleanroom
- Validation of particle counter and cleanroom
- Integrate 6 DOF robot arm



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Questions?

Thank you!